Sri Vasavi Engineering College
(Autonomous)

Computer Science & Engineering (Artificial Intelligence)

Major Project Review-I

Batch No: 10

Project Title: RoadGuard : An Explainable Deep Learning-Based Road Damage Assessment and Intelligent Maintenance Prioritization System

S.No	Reg.No	Student Name
1	23A81A4383	ChSatyanarayana
2	23A81A4394	G SSPriya
3	23A81A4373	AJ S Prasanna
4	24A85A4309	G Nikhil

Project Guide
Mrs. G Venkata Lakshmi

Scanned with OKEN Scanner

Abstract

RoadGuard is an AI-based system designed to detect and assess road damage and help authorities plan road maintenance. It uses images or videos captured from mobile phones or vehicle-mounted cameras to identify road defects such as potholes, cracks, and surface damage using deep learning.

The system estimates the severity of each defect based on its type, size, and density. GPS is used to identify the damaged road location, road segment, and responsible authority. A priority score is then generated based on the severity of damage, road importance, location risk, and repeated reports.

The detected information is stored in a database and displayed through an interactive map and authority dashboard. RoadGuard supports both user complaints and automatic road inspections, helping to identify previously unreported damage. The system provides a data-driven approach to improve road monitoring, maintenance planning, and road safety.

Scanned with OKEN Scanner

Outline

1. Introduction & Motivation
2. Problem Statement
3. Literature Review
4. Research Gap
5. Objectives & Questions
6. Methodology
7. Progress So Far
8. Timeline Plan
9. Expected Contributions
10. References

Scanned with OKEN Scanner

Background & Motivation

- Roads are important for safe and efficient transportation, but they often develop potholes, cracks, and surface damage due to traffic, weather, and aging.
- Manual road inspection requires a lot of time and human effort, especially when monitoring large road networks.Damaged roads are not always reported immediately, which can lead to safety risks, vehicle damage, and increased repair costs.
- Existing road-damage detection methods mainly focus on identifying defects, but may not provide complete information for maintenance planning.
- There is a need for an automated and intelligent system that can detect road damage, estimate its severity, locate the damage, and identify the responsible authority.
- RoadGuard is proposed to combine deep learning, GPS-based location, severity assessment, and maintenance prioritization to support faster and data-driven road maintenance decisions.

Scanned with OKEN Scanner

Problem Statement

- **Core problem:** Existing road inspection methods are time-consuming and often rely on manual identification of road damage.
- **Context:** Large road networks contain different types of damage such as potholes, longitudinal cracks, transverse cracks, alligator cracks, and surface deterioration.
- **Key challenges:**
  - Difficulty in detecting road damage accurately and consistently
  - Lack of automated severity assessment for detected defects
  - Difficulty in identifying the exact road location and responsible authority
  - Lack of intelligent prioritization for maintenance activities
- Existing approaches mainly focus on road-damage detection and do not jointly provide severity assessment, GPS-based mapping, authority identification, and maintenance prioritization in one integrated framework.

Scanned with OKEN Scanner

Literature Review: Scope & Search Strategy

- Databases: IEEE Xplore, Scopus, Web of Science, ACM DL, ScienceDirect, arXiv
- Keywords: "road damage detection", "pothole detection deep learning", "pavement distress detection", "road crack detection CNN", "road damage severity assessment", "intelligent road maintenance"
- Time window: 2020–2026
- Inclusion: Peer-reviewed works on road damage detection, pavement distress analysis, deep learning-based detection or segmentation, road damage datasets, and intelligent maintenance systems
- Exclusion: Studies unrelated to road infrastructure, non-image-based approaches without relevance to automated road inspection, duplicate works, and non-English works

Scanned with OKEN Scanner

Key Themes in Existing Work

Year	Work	Method	Dataset	Key Limitation
2020	Automated Crack Segmentation	Thresholding + Canny + Level Set	Pavement crack images	Limited performance in complex scenes
2022	Pavement Crack Detection and Classification	LBP + PCA + SVM	Pavement crack images	Requires handcrafted features
2022	Automatic Pavement Damage Prediction	Machine Learning Algorithms	Pavement damage data	Limited generalization to new environments
2024	Automated Pavement Distress Detection	DARNet + Attention + Feature Fusion	LNTU_RDD_GS	Focuses mainly on distress segmentation
2024	Pavement Distress Detection Comparison	U-Net / DeepLabv3+ / PSPNet / HRNet	LNTU_RDD_GS	Small cracks and complex defects may be missed
2024	DARNet Evaluation on Public Datasets	DARNet + CFASPP + DAAM + REM + WCEL	Crack500, CFD, GAPs384, Aigle RN	Does not address maintenance prioritization

Scanned with OKEN Scanner

Strengths & Limitations of Existing Approaches

- **Strengths:**
  - Deep learning methods provide more accurate and automated road-damage detection than traditional manual and image-processing methods.
  - CNN-based segmentation and attention mechanisms improve the detection of cracks, potholes, and other pavement distress.
  - Public datasets such as Crack500, CFD, GAPs384, and LNTU_RDD_GS support training and evaluation of road-damage detection models.
- **Limitations:**
  - Small cracks and different-sized pavement defects can be difficult to detect accurately.
  - Complex backgrounds, stains, shadows, uneven illumination, and image noise can cause missed or incorrect detections.
  - Existing approaches mainly focus on defect detection or segmentation and provide limited support for severity assessment and maintenance planning.
- Existing approaches rarely provide an end-to-end framework that combines road-damage detection, severity assessment, GPS-based mapping, authority identification, and intelligent maintenance prioritization.

Scanned with OKEN Scanner

Identified Research Gap

- No unified AI-based road assessment framework that simultaneously:
  1. Automatically detects and classifies multiple types of road damage.
  2. Estimates damage severity and prioritizes road maintenance.
- Existing deep learning approaches mainly focus on road-damage detection or segmentation and provide limited integration with location-based road assessment.
- Current systems do not sufficiently combine GPS-based localization, road-authority identification, severity assessment, and maintenance prioritization.
- **Gap type:** Methodological + Practical.

Scanned with OKEN Scanner

From Gap to Opportunity

- Addressing the gap enables automated road-damage assessment without relying only on costly and time-consuming manual inspections.
- Combining damage detection, severity assessment, GPS localization, and maintenance prioritization can improve decision-making and help authorities respond to critical road damage faster.
- **Opportunity:** Design an end-to-end framework that uses deep learning and location-based information for intelligent road assessment and maintenance prioritization.

Scanned with OKEN Scanner

Research Objectives

- **O1:** Develop a deep learning-based system to automatically detect and classify different types of road damage from images or videos.
- **O2:** Estimate the severity of detected road damage based on defect type, size, and concentration.
- **O3:** Integrate GPS-based localization, road-segment mapping, and authority identification with damage assessment.
- **O4:** Develop an intelligent maintenance prioritization system that ranks damaged road segments based on severity, road importance, location risk, and repeated reports.

Scanned with OKEN Scanner

Research Questions / Hypotheses

- **RQ1:** Can a deep learning-based model accurately detect and classify different types of road damage from real-world road images or videos?
- **RQ2:** How does automated severity assessment affect the identification and prioritization of critical road damage?
- **RQ3:** How effectively can GPS-based localization, road-segment mapping, and authority identification support road maintenance planning?
- **Hypothesis:** An integrated road assessment framework combining damage detection, severity estimation, GPS localization, and maintenance prioritization will provide more effective and actionable road monitoring than detection-only approaches.

Scanned with OKEN Scanner

Overall Methodology Overview

- **High-level pipeline:**
  1. Road image/video collection from mobile devices or vehicle-mounted cameras
  2. Image preprocessing and road-damage feature extraction
  3. Deep learning-based road damage detection and classification
  4. Severity estimation and GPS-based road localization
  5. Road-segment mapping, authority identification, and maintenance prioritization
  6. Database storage and visualization through map and authority dashboard
- **Main modules:** Data Collection Module, Damage Detection Engine, Severity Assessment Module, GPS & Mapping Module, Priority Engine, Dashboard.
- **Integrated approach:** Deep learning + GPS-based mapping + database management + intelligent maintenance prioritization.

Scanned with OKEN Scanner

Proposed Approach / System Design

- **Core idea:** An end-to-end road assessment framework that automatically detects road damage, estimates its severity, identifies its location and responsible authority, and prioritizes maintenance actions.
- **Differentiators:**
  - Combines deep learning-based damage detection with severity assessment and GPS-based road localization.
  - Integrates road-segment mapping and authority identification for actionable maintenance planning.
  - Uses an intelligent priority score to rank damaged road segments based on severity, road importance, location risk, and repeated reports.
- **Key design choices:** Deep learning object detection, image/video preprocessing, GPS-based mapping, database management, and rule-based maintenance prioritization.

Scanned with OKEN Scanner

Methodological Details

- **Data:** Road images and videos collected using mobile devices or vehicle-mounted cameras. Data includes potholes, longitudinal cracks, transverse cracks, alligator cracks, and surface damage, along with GPS location information.
- **Models:** Deep learning-based object detection model for road-damage detection and classification, combined with severity assessment and GPS-based road mapping.
- **Baselines:** Traditional image-processing methods, machine learning-based approaches, and existing deep learning road-damage detection and segmentation models.
- **Metrics:**
  - Detection performance: Accuracy, Precision, Recall, F1-score, and IoU
  - Severity assessment: Severity classification accuracy
  - System performance: Processing time and computational efficiency
  - Maintenance prioritization: Correct ranking of high-risk road segments

Scanned with OKEN Scanner

Evaluation Plan

- **Setup:** Python-based deep learning environment, road-damage datasets, GPU-enabled system, database, GPS mapping, and web-based dashboard.
- **Scenarios:** Different road conditions, damage types, image qualities, lighting conditions, and road environments.
- **Experiments:** Evaluation of detection accuracy, severity assessment, GPS localization, maintenance prioritization, scalability, and comparison with existing approaches.
- **Validation:** Multiple test samples, performance metrics, error analysis, and comparison of predicted results with ground-truth road-damage annotations.

Scanned with OKEN Scanner

Work Completed

- Comprehensive literature survey completed on road-damage detection, pavement distress assessment, and deep learning approaches.
- Problem statement and research gap refined based on existing road-damage detection approaches.
- Initial RoadGuard system architecture and end-to-end workflow designed.
- Road-damage categories, severity assessment approach, GPS localization, and maintenance prioritization workflow defined.
- Initial technology stack identified, including Python, deep learning frameworks, database, GPS mapping, and web-based dashboard development.

Scanned with OKEN Scanner

Preliminary Results

- Initial testing of the road-damage detection pipeline shows promising results in identifying common defects such as potholes and cracks.
- The detection approach is expected to improve automated road inspection compared with fully manual inspection methods.
- Key challenges include detecting small cracks, handling shadows and uneven lighting, and distinguishing road damage from background noise.
- These challenges are being addressed through image preprocessing, deep learning-based feature extraction, and improved training with diverse road images.

Scanned with OKEN Scanner

Proposed Timeline

- **Month 1–2:** Finalize problem definition, literature review, research gap, system requirements, and road-damage dataset collection.
- **Month 3–4:** Prepare and preprocess road-damage images and implement the YOLO-based road damage detection model for potholes, cracks, and surface damage.
- **Month 5–6:** Develop the severity estimation module, GPS-based location detection, road mapping, and municipal authority identification.
- **Month 7–8:** Develop the Django-based user reporting system and municipal officer dashboard, including complaint management, interactive maps, and maintenance priority scoring.
- **Month 9–10:** Integrate all modules, perform model evaluation, system testing, accuracy analysis, duplicate-report detection, and robustness studies.
- **Month 11+:** Optimize the system, complete documentation, prepare the research paper and thesis, and finalize the project deployment.

Scanned with OKEN Scanner

Next Steps Before Second Review

- Complete road-damage dataset collection, preprocessing, and annotation for potholes, cracks, and surface damage.
- Train and evaluate the initial YOLO-based road-damage detection model and obtain baseline detection results.
- Develop the severity estimation and GPS-based road-damage location modules.
- Implement the initial Django-based citizen reporting system and municipal officer dashboard.
- Prepare the methodology, system architecture, and initial-results sections of the report/paper.

Scanned with OKEN Scanner

Anticipated Contributions

- An AI-based road damage assessment framework that automatically detects potholes, cracks, and surface defects from user-captured road images.
- An intelligent severity and maintenance prioritization mechanism based on damage type, size, defect density, road importance, location risk, and repeated reports.
- A GPS-enabled geospatial system for identifying damaged road locations and mapping them to the responsible municipal authority.
- A citizen reporting platform and municipal dashboard for complaint submission, monitoring, repair assignment, and maintenance status tracking.
- A unified system integrating computer vision, geospatial mapping, explainable AI, and intelligent maintenance prioritization for efficient road-condition monitoring.

Scanned with OKEN Scanner

Key References

- Maeda et al., "Road Damage Detection and Classification Using Deep Neural Networks with Smartphone Images," *Computer-Aided Civil and Infrastructure Engineering*, 2018.
- Arya et al., "Deep Learning-Based Road Damage Detection and Classification for Multiple Road Types," *Computer-Aided Civil and Infrastructure Engineering*, 2021.
- "RDD2022: A Multi-National Image Dataset for Automatic Road Damage Detection," *International Journal of Pavement Engineering*, 2022.
- "Road Damage Detection Using Deep Learning-Based Object Detection Models," *IEEE/International Conference Proceedings*, 2023.
- Bochkovskiy, A., Wang, C.-Y., and Liao, H.-Y. M., "YOLOv4: Optimal Speed and Accuracy of Object Detection," 2020.
- Wang, C.-Y., Bochkovskiy, A., and Liao, H.-Y. M., "YOLOv7: Trainable Bag-of-Freebies Sets New State-of-the-Art for Real-Time Object Detectors," *CVPR*, 2023.
- "Deep Learning-Based Pothole Detection for Road Infrastructure Monitoring," *Journal of Transportation Engineering*, 2023.
- "Artificial Intelligence and Computer Vision for Automated Pavement Condition Assessment," *Automation in Construction*, 2024.

Scanned with OKEN Scanner

Thank You

Questions & Discussion

Students: Challa Satyanarayana , Grandhi Sri Swarna Priya , Anisetti Janaki Sai Prasanna, Gollakoti Nikhil

Guide: Mrs. G Venkata Lakshmi

Sri Vasavi Engineering College, Tadepalligudem, Andhra Pradesh

Scanned with OKEN Scanner